

Proto-Indo-European Laryngeals and Vowel Coloring in Anatolian and Greek

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Abstract

This working paper investigates the reflexes of the three Proto-Indo-European laryngeal phonemes ($*h_1$, $*h_2$, $*h_3$) in Anatolian (Hittite) and archaic Greek. We examine the conditions under which laryngeals produce vocalic coloring versus uncolored preservation, paying special attention to word-initial positions and root ablaut grades ($*e$ -grade versus $*o$ -grade). We demonstrate that the Anatolian evidence provides conclusive confirmation of Ferdinand de Saussure’s original *coefficients sonantiques*.

1 Introduction

The laryngeal theory, first proposed by Ferdinand de Saussure in 1879 as abstract algebraic entities in the PIE root system, was vindicated following Jerzy Kurylowicz’s identification of Hittite $\check{h}$ with Saussure’s hypothetical coefficients.

In traditional Indo-European phonology, three distinct laryngeals are reconstructed:

1. $*h_1$: Neutral laryngeal (often assumed to be a glottal stop [ʔ] or voiceless glottal fricative [h]), leaving adjacent $*e$ uncolored.
2. $*h_2$: a -coloring laryngeal (reconstructed as a voiceless uvular or pharyngeal fricative [χ] or [ħ]), shifting adjacent $*e > *a$.
3. $*h_3$: o -coloring laryngeal (reconstructed as a voiced labialized velar or pharyngeal [ɣ^w] or [ʕ^w]), shifting adjacent $*e > *o$.

2 Phonetic Reflexes

2.1 Anatolian Reflexes

Hittite directly preserves $*h_2$ and $*h_3$ in initial and intervocalic positions as $\check{h}$ and $\check{h}\check{h}$, while $*h_1$ is typically lost:

- PIE $*h_2enti$ “in front, opposite” → Hittite $hānti$ “in front”, Greek $ánti$, Latin *ante*.
- PIE $*h_3ep-$ “to take” → Hittite *happ-* “to join, hand over”, Latin *apiscor*.

3 Conclusion & Future Work

Computational modeling of phonetic change networks allows us to rigorously verify whether reconstructed laryngeal phonemes maintain phonetic plausibility under cross-linguistic acoustic constraints.